

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: *Perform dry run testing to trace the execution of algorithms, analyze intermediate steps, and verify the correctness of processes. (BL3)*

Assessment Tool Weightage: 40%

Duration : 45 Minutes

QUESTIONS: 2

Total Marks:20

Annexure A

DRY RUN TEST

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question 1: Dry Run of A* Search Algorithm (10 Marks)

Formula: $f(n) = g(n) + h(n)$, where $g(n)$ is the path cost from A and $h(n)$ is the heuristic value.

Node	$h(n)$
A	7
B	6
C	4
D	3
E	2
G	0

Iteration 1 – Current Node: A

- $g(A) = 0$, $h(A) = 7$, $f(A) = 7$
- Closed List = {A}
- B: $g = 2$, $h = 6$, $f = 8$
- C: $g = 4$, $h = 4$, $f = 8$
- Open List = {B(8), C(8)}
- Choose B.

Iteration 2 – Current Node: B

- $g(B) = 2$, $h(B) = 6$, $f(B) = 8$
- Closed List = {A, B}
- C through B: $g = 5$, which is worse than existing $g(C) = 4$.
- D: $g = 2 + 7 = 9$, $f(D) = 9 + 3 = 12$
- E: $g = 2 + 2 = 4$, $f(E) = 4 + 2 = 6$
- Open List = {E(6), C(8), D(12)}
- Choose E.

Iteration 3 – Current Node: E

- $g(E) = 4$, $h(E) = 2$, $f(E) = 6$
- Closed List = {A, B, E}
- D through E: $g = 4 + 2 = 6$, $f(D) = 6 + 3 = 9$. Update D.
- C through E: $g = 4 + 3 = 7$, worse than existing $g(C) = 4$.
- Open List = {C(8), D(9)}
- Choose C.

Iteration 4 – Current Node: C

- $g(C) = 4$, $h(C) = 4$, $f(C) = 8$
- Closed List = {A, B, E, C}
- No better path is obtained from C.
- Open List = {D(9)}
- Choose D.

Iteration 5 – Current Node: D

- $g(D) = 6, h(D) = 3, f(D) = 9$
- Closed List = {A, B, E, C, D}
- G: $g(G) = 6 + 2 = 8, h(G) = 0, f(G) = 8$
- Open List = {G(8)}
- Choose G.

Iteration 6 – Current Node: G

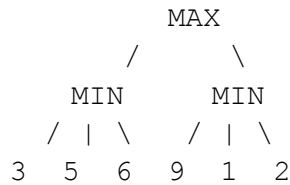
- $g(G) = 8, h(G) = 0, f(G) = 8$
- Closed List = {A, B, E, C, D, G}
- G is the goal node. Stop.

Final Answer

- **Optimal Path: A $\rightarrow$ B $\rightarrow$ E $\rightarrow$ D $\rightarrow$ G**
- **Total Path Cost = 2 + 2 + 2 + 2 = 8**

Question 2: Dry Run of Minimax with Alpha-Beta Pruning (10 Marks)

Game Tree:



Step 1 – Root MAX

- Initially $\alpha = -\infty$ and $\beta = +\infty$.
- Evaluate the left MIN node first.

Step 2 – Left MIN Node

- Leaf 3: MIN value = 3, $\beta = 3$.
- Leaf 5: $\min(3, 5) = 3$.
- Leaf 6: $\min(3, 6) = 3$.
- Therefore $\text{MIN}(3, 5, 6) = 3$.
- Return 3 to MAX. Now $\alpha = \max(-\infty, 3) = 3$.

Step 3 – Right MIN Node

- MAX passes $\alpha = 3$ and $\beta = +\infty$.
- Leaf 9: MIN value = 9, $\beta = 9$.
- Leaf 1: MIN value = 1, $\beta = 1$.
- Now $\beta \leq \alpha$, because $1 \leq 3$. Therefore an Alpha-Beta cutoff occurs.
- The remaining leaf 2 is pruned.
- Right MIN returns 1.

Step 4 – Root MAX Decision

- MAX compares $\max(3, 1) = 3$.
- Therefore MAX chooses the left subtree.

Final Answers

- Best move for MAX: Left branch / Left MIN subtree**
- Final Minimax Value: 3**
- Pruned Node: 2**
- Final Result: MAX chooses the left subtree with minimax value 3.**